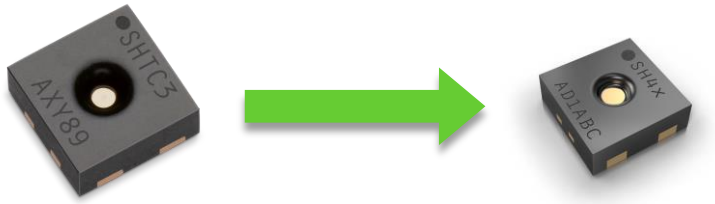
# SHTC3 – SHT4x Transition Guide

Boarding the new flagship RH/T sensor generation

SHTC3 SHT4x




## Features

- Improved accuracy, power consumption, and robustness
- Powerful internal heater for self-decontamination
- Superior versatility and technology from two decades of sensor development
- Relative humidity accuracy: up to ±1.0 %RH
- Temperature accuracy: up to ±0.1 °C
- Supply voltage: 1.08 V … 5.5 V
- Average current: 0.4 µA (at meas. rate 1 Hz)
- Idle current: 80 nA
- Fully functional in condensing environment

## General Description

Once introduced as a specialty digital humidity (RH) and temperature (T) sensor for high volume consumer electronics, the SHTC3 enabled outstanding sensing performance in a small package for several years. Sensirion now proudly recommends it's all new flagship successor from the SHT4x family, which profits from about two decades of RH/T sensor development. Dedicated to best-in-class performance, low power consumption, smallest footprint, and attractive pricing, our new SHT4x sensors are the products of choice for many SHTC3 applications. In particular, the SHT4x outperforms the SHTC3 in every aspect and offers versatile add-ons, such as a powerful heater for self-decontamination, conformal coating protection, or filter membranes, previously not available for the SHTC3.

## Important Changes

| Parameter | SHTC3 | SHT4x |
|---|---|---|
| Dimensions ($mm^3$) | 2.0 × 2.0 × 0.75 | 1.5 × 1.5 × 0.5 |
| Pin assignment | Comparable | |
| Interface | $I^2C$, single address | $I^2C$, multiple addresses |
| Supply voltage (V) | 1.62 – 3.6 | 1.08 – 5.5[1] |
| Av. current (µA @ 1Hz) | 4.9 | 0.6 |
| Typ. RH accuracy (%RH) | ±2.0 | ±1.8 – ±1.0 |
| Typ. T accuracy (°C) | ±0.2 | ±0.2 – ±0.1 |
| Response time $\tau$ 63% (s) | 8 | 4 |
| Filter membrane | Not available | Available |
| Conformal coating | Not possible | Possible |
| Additional features | - | Powerful heater with $\Delta T \geq 60°C$, |

[1] SHT4xI has a higher possible supply voltage up to 5.5V

| | | |
|---|---|---|
| | | Full condensation robustness |

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# 1 Purpose

This document aims to provide a high-level guideline to replace SHTC3 with sensors from the SHT4x family and outlines important differences to be considered in design-in processes.

# 2 Performance Comparison

## 2.1 Relative Humidity and Temperature

| Parameter | Conditions | SHTC3 | SHT4x | Units |
|---|---|---|---|---|
| **Relative humidity** | | | | |
| RH accuracy[2] | Typ. | **±2** | **1.0 … ±1.8** | %RH |
| High repeatability[3] | - | 0.1 | 0.08 | %RH |
| Resolution[4] | - | 0.01 | 0.01 | %RH |
| Hysteresis | - | ±1 | ±1 | %RH |
| Specified range[5] | extended[6] | 0 – 100 | 0 – 100 | %RH |
| Response time[7] | $\tau$ 63% | **8** | **4** | s |
| Long-term drift[8] | Typ. | < 0.25 | < 0.25 | %RH/y |
| Condensation behavior | Droplet formation | **Slight signal drop** | **No signal drop** | - |
| **Temperature** | | | | |
| T Accuracy[1] | Typ. | **±0.2** | **±0.1** | °C |
| High repeatability[2] | - | **0.1** | **0.04** | °C |
| Resolution[3] | - | 0.01 | 0.01 | °C |
| Specified range[4] | - | −40 – +125 | −40 – +125 | °C |
| Response time[9] | $\tau$ 63% | **< 5 – 30** | **2** | s |
| Long-term drift[10] | Typ. | < 0.02 | < 0.01, < 0.03 | °C/y |

**Table 1.** Humidity and temperature specifications of the SHTC3 and SHT4x, where bold values highlight important differences. For further details, kindly refer to the SHTC3 and SHT4x datasheets.

[2] For definition of typ. accuracy, please refer to the document “Sensirion Humidity Sensor Specification Statement”.

[3] The stated repeatability is 3 times the standard deviation (3σ) of multiple consecutive measurement values at constant conditions and is a measure for the noise on the physical sensor output. Different repeatability commands are listed in **Table 5**.

[4] Resolution of A/D converter.

[5] Specified range refers to the range for which the humidity or temperature sensor specification is guaranteed.

[6] For details about recommended humidity and temperature operating range, please refer to the SHT4x Datasheet.

[7] Time for achieving 63% of a humidity step function, valid at 25°C and 1 m/s airflow. Humidity response time in the application depends on the design-in of the sensor.

[8] Typical value for operation in normal RH/T operating range. Max. value is < 0.5 %RH/y. Value may be higher in environments with vaporized solvents, out-gassing tapes, adhesives, packaging materials, etc. For more details, please refer to Handling Instructions.

[9] Temperature response time depends on heat conductivity of sensor substrate and design-in of sensor in application.

[10] Max. value is < 0.04°C/y. The long-term drift for SHT43 is < 0.01 C/y.

### 2.2 Electrical Characteristics

| Parameter | Symbol | Conditions | SHTC3 | | | SHT4x | | | Units |
|---|---|---|---|---|---|---|---|---|---|
| | | | Min. | Typ. | Max. | Min. | Typ. | Max. | |
| Supply voltage | $V_{DD}$ | | **1.62** | **3.3** | **3.6** | **1.08** | **3.3** | **3.6** | V |
| Power-up/down level | $V_{POR}$ | Static power supply | **1.28** | **1.4** | **1.55** | **0.6** | **-** | **1.08** | V |
| Supply current (heater not activated) | $I_{DD}$ | Sleep/Idle | - | **0.3** | **0.6** | - | **0.08** | - | µA |
| | | Meas. | - | **430** | **900** | - | **320** | - | µA |
| | | Average | - | **4.9** | - | - | **0.4 (low)<br>2.2 (high)** | - | µA |
| Power consumption | - | Average | **16.2** | - | - | - | **0.5 (low)<br>2.6 (high)** | - | µW |
| Low level input voltage | $V_{IL}$ | - | - | - | **0.42 $V_{DD}$** | 0 | - | **0.3 $V_{DD}$** | V |
| High level input voltage | $V_{IH}$ | - | 0.7 $V_{DD}$ | - | - | 0.7 $V_{DD}$ | - | **$V_{DD}$** | V |
| Application circuit design | - | - | Identical, for details see SHTC3 or SHT4x datasheet | | | | | | - |

**Table 2.** Key electrical specifications of the SHTC3 and SHT4x, where bold values highlight important differences. For further details, kindly refer to the SHTC3 and SHT4x datasheets.

### 2.3 Timing Specifications

| Parameter | Symbol | Conditions | SHTC3 | | | SHT4x | | | Units |
|---|---|---|---|---|---|---|---|---|---|
| | | | Min | Typ. | Max | Min | Typ. | Max | |
| Power-up time | $t_{PU}$ | After hard reset, $V_{DD} \geq V_{POR}$ | - | **0.18** | **0.24** | - | - | **1** | ms |
| Soft reset time | $t_{SR}$ | After soft reset | - | **0.18** | **0.24** | - | - | **1** | ms |
| Measurement duration | $t_{Meas}$ | Medium repeatability | - | **10.8** | **12.1** | - | **3.7** | **4.5** | ms |
| Heater-on duration | $t_{Heater}$ | | - | - | - | 0.1 | - | 1.0 | s |

**Table 3.** Key timing specifications of the SHTC3 and SHT4x, where bold values highlight important differences. For further details, kindly refer to the SHTC3 and SHT4x datasheets.

## 3 Flagship SHT4x Feature: Built-In Heater

The SHT4x sensor incorporates a powerful on-chip heater, which can be used for self-decontamination, *e.g.* in environments with solvents present, and periodical creep compensation in prolonged application in highest humidity. It provides an over-temperature of about 60 °C and can be switched on by the command specified in **Table 5**, after which the heater will run for 1 second. After 1 second, a temperature and humidity measurement is started and the heater will be automatically turned off after the measurement is finished. This safety feature prevents permanent turn-on of the heater. There is no dedicated command to turn off the heater. If higher temperatures than achievable by heating for 1 second are desired, consecutive heating commands need to be sent to the sensor.

## 4 Package Design Differences

The SHT4x comes in a new open-cavity dual flat no lead (DFN) package design in order to enable significant size reduction while allowing additional features, such as conformal coating, protection cover, and filter membrane compatibility. Similar to the SHTC3, the humidity sensor opening is centered on the top side of the SHT4x package (see **Figures 1,2**). The bottom side of the both DFN package exposes metallic contacts, which are Ni/Pd/Au coated, while the side walls of the contacts are bare copper.

| Parameter | Units | SHTC3 | SHT4x | Comment |
|---|---|---|---|---|
| Size | mm | 2.0 x 2.0 x 0.75 | 1.5 x 1.5 x 0.5 | For details, see Figures 1,2. |
| Sensor opening | - | Top | Top | |
| Protection compatibility | - | none | Compatible with conformal coating and filter membranes | |
| Pin Layout | - | 2 x 2 pins | 2 x 2 pins | |
| Pin Assignment | - | VDD SHTC3 VSS<br>SCL AXY89 SDA | SDA SHT4 VSS<br>SCL SABC VDD | Drawings not to scale.<br>VDD: Supply voltage<br>SCL: Serial clock<br>SDA: Serial data bidirectional<br>VSS: Ground |
| Pin Size | mm | 0.35 x 0.35 | 0.3 x 0.3 | |
| Pin Pitch | mm | 1.0 | 0.8 | |
| Pin Material | - | Ni/Pd/Au coated Cu | Ni/Pd/Au coated Cu | |
| Housing Material | - | Epoxy housing | Epoxy housing | |

**Table 4.** Key package differences between the SHTC3 and SHT4x. For further details, kindly refer to the SHTC3 and SHT4x datasheets.

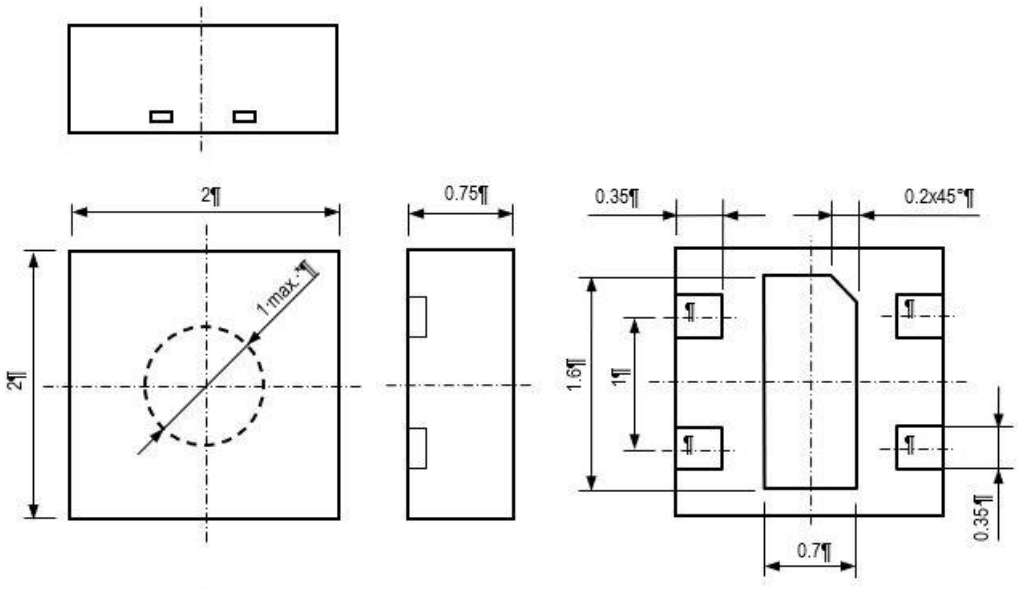



**Figure 1.** Dimensional drawing of the SHTC3 (units mm).

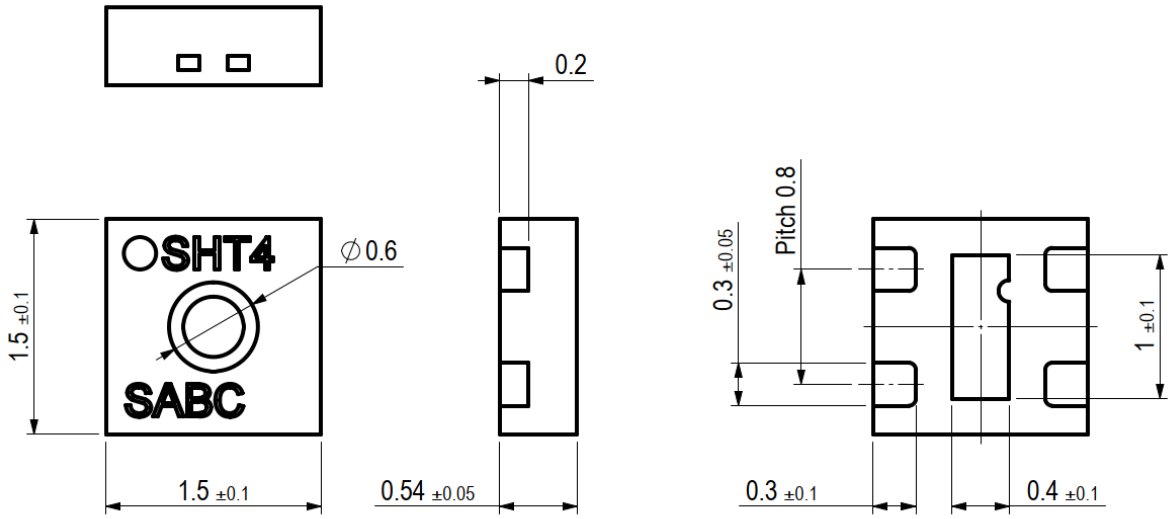



**Figure 2.** Dimensional drawing of the SHT4x including package tolerances (units mm).

## 5 Communication Compatibility

Both chips feature the I²C communication protocol, however, while the SHTC3 has a fixed I²C address, the SHT4x features a default I²C address (0x44) and alternative addresses for high flexibility in all applications. Addressing a specific SHT4x sensor is done by sending its 7-bit I²C address followed by an eighth bit, denoting the communication direction: "Zero" indicates transmission to the sensor, *i.e.* "write", a "one" indicates a "read" request.

In addition, the SHT4x features different measurement options for different precision needs and a heater option, as detailed in **Table 5** and **Section 3**.

| Command | | Description |
|---|---|---|
| BIN | HEX | |
| 1111 1101 | FD | Measure T & RH with highest precision (high repeatability) |
| 1111 0110 | F6 | Measure T & RH with medium precision (medium repeatability) |
| 1110 0000 | E0 | Measure T & RH with lowest precision (low repeatability) |
| 1000 1001 | 89 | Read serial |
| 1001 0100 | 94 | Soft Reset |
| 0011 1001 | 39 | Activate highest heater power for 1s |

**Table 5.** Short overview of I²C commands for the SHT4x.

For further details on the I²C communication, such as general protocol description, data types and lengths, and checksum calculation, kindly refer to the SHT4x datasheet.

## 6 Quality and Material Contents

Qualification of the SHTC3 and SHT4x is performed based on the JEDEC JESD47 qualification test method. While both devices are fully RoHS and REACH compliant, the SHT4x is also WEEE compliant.

## 7 Further Information

This transition guide aims at providing an overview of the key differences between the SHTC3 and the SHT4x, yet it might not be fully inclusive. For further reading on the SHT4x specifications, communication, operation, and application, please consult the dedicated SHTC3 and SHT4x documents provided on the Sensirion webpage www.sensirion.com. In case you are in need of specific details, or would like to request assistance in transitioning from the SHTC3 to the SHT4x or any other Sensirion product, please consult us directly at www.sensirion.com/en/about-us/contact/.

## 8 Revision History

| Date | Version | Page(s) | Changes |
|---|---|---|---|
| April 2021 | 1 | all | Initial release |
| November 2023 | 1.1 | All | Updated SHT4x information |

# Important Notices

**Warning, Personal Injury**

**Do not use this product as safety or emergency stop devices or in any other application where failure of the product could result in personal injury. Do not use this product for applications other than its intended and authorized use. Before installing, handling, using or servicing this product, please consult the data sheet and application notes. Failure to comply with these instructions could result in death or serious injury.**

If the Buyer shall purchase or use SENSIRION products for any unintended or unauthorized application, Buyer shall defend, indemnify and hold harmless SENSIRION and its officers, employees, subsidiaries, affiliates and distributors against all claims, costs, damages and expenses, and reasonable attorney fees arising out of, directly or indirectly, any claim of personal injury or death associated with such unintended or unauthorized use, even if SENSIRION shall be allegedly negligent with respect to the design or the manufacture of the product.

**ESD Precautions**

The inherent design of this component causes it to be sensitive to electrostatic discharge (ESD). To prevent ESD-induced damage and/or degradation, take customary and statutory ESD precautions when handling this product.
See application note "ESD, Latchup and EMC" for more information.

**Warranty**

SENSIRION warrants solely to the original purchaser of this product for a period of 12 months (one year) from the date of delivery that this product shall be of the quality, material and workmanship defined in SENSIRION's published specifications of the product. Within such period, if proven to be defective, SENSIRION shall repair and/or replace this product, in SENSIRION's discretion, free of charge to the Buyer, provided that:

- notice in writing describing the defects shall be given to SENSIRION within fourteen (14) days after their appearance;
- such defects shall be found, to SENSIRION's reasonable satisfaction, to have arisen from SENSIRION's faulty design, material, or workmanship;
- the defective product shall be returned to SENSIRION's factory at the Buyer's expense; and
- the warranty period for any repaired or replaced product shall be limited to the unexpired portion of the original period.

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